

Sarh

صَرَح

A national platform for Libya unifying the Real Estate Registry and Digital Identity issuance — built on verifiable credentials, append-only audit, and tamper-evident deeds.

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Introduction

1.1 Problem Statement

Libya's real estate sector currently relies on paper deeds issued, transferred, and verified manually across municipal offices. The lack of a central registry has produced three persistent problems: (i) *title insecurity*, where multiple owners may legitimately hold conflicting paper deeds for the same parcel; (ii) *verification opacity*, which forces buyers, banks, and courts to rely on uncertain provenance chains; and (iii) *identity fragility*, where there is no national digital identity that can be cryptographically bound to a property holding.

These problems are amplified by the country's historical disruptions: archive losses, multiple parallel administrative systems, and the absence of a canonical reference dataset for parcel geometry.

1.2 Research Questions

1. How can a digital registry coexist with — rather than replace — legitimate paper deeds during a multi-year transition?
2. What architecture supports a future national digital identity (when issued by the Libyan state) without forcing a data migration?
3. How can deeds be made tamper-evident and machine-verifiable by any third party (bank, court, citizen)?
4. How can polygon overlap conflicts be surfaced for officer review without blocking submissions?

1.3 Objectives

- **01.** Build a unified web + mobile platform that registers properties, issues digital ID cards (NTAG 424 DNA), and produces tamper-evident PAdES-signed PDF deeds.
- **02.** Preserve every paper-era artifact via a `legacy_national_no` field on every citizen and a deed reference field on every property.
- **03.** Provide a public verification page accessible to anyone with a deed code or QR scan, requiring no login.
- **04.** Enforce an append-only audit log, defended at the database level by `INSTEAD OF UPDATE/DELETE` triggers.

- **05.** Deliver an Arabic-first RTL experience across all citizen-facing surfaces, with Latin reserved for codes and identifiers.

1.4 Literature Review

Comparable national-registry initiatives provide instructive prior art. The table below contrasts Sarh's positioning against three reference systems.

SYSTEM	COUNTRY	IDENTITY MODEL	DEED MODEL	PUBLIC VERIFY
e-Estonia	Estonia	National eID (chip card)	X-Road federated	Yes (X-Road)
Bhumi	India (West Bengal)	Aadhaar bind	Centralized DB	Limited
SR Lanka Land Registry	Sri Lanka	NIC bind	PDF + signature	Yes
Sarh	Libya	NTAG 424 DNA + SSI (ACA-Py)	PAdES + QR + SHA-256	Yes (no login)

1.5 Scope

IN SCOPE

- Citizen, officer, ID-issuer, admin, and verifier portals
- Mobile app (Flutter, Riverpod) for citizens
- NFC-card encoding and SUN verification
- PAdES-signed deeds with QR
- SSI verifiable credentials (did:sov via ACA-Py)
- Public deed verification page

OUT OF SCOPE (THIS VERSION)

- Direct integration with bank settlement systems
- Court e-filing integration
- Cadastral surveys (we accept geometry as input)
- Mortgage workflows
- Cross-border identity federation

Methodology

2.1 Development Process

Sarh follows an **incremental SDLC** partitioned into thirteen delivery phases (0 through 12). Each phase has an explicit Definition of Done: code passes lint and unit tests, an end-to-end happy-path runs green, API endpoints appear in OpenAPI, permissions are reviewed against the role-permission map, the Arabic UI is sanity-checked on RTL with real Arabic content, and audit log entries are verified for every write path.

2.2 Feasibility Analysis

TECHNICAL

All major components are commodity open-source: ASP.NET Core 8 for the API, SQL Server 2019/2022 for the data store, Angular 21 + Flutter for clients, Hyperledger ACA-Py for SSI, and NTAG 424 DNA chips (Mifare ecosystem) for cards. No exotic vendor lock-in.

OPERATIONAL

The reference deployment is a single Docker Compose stack on a VPS (api, web, mssql, aca-py, nginx, clamav). Backups are nightly full + hourly log-shipping for SQL Server.

ECONOMIC

Per-card cost is approximately \$0.85 (NTAG 424 DNA in volume). The expensive moving parts are operator training and the survey of legacy paper deeds — not technology spend.

2.3 Risk Register

RISK	LIKELIHOOD	IMPACT	MITIGATION
NFC card cloning	Medium	Critical	NTAG 424 DNA SUN messages with rolling counter; static UID is never trusted alone
Polygon overlap with legitimate paper deed	High	Medium	Soft-block: warn officer, do not auto-reject; resolve via review committee
National eID launch forces migration	Medium	High	<code>legacy_national_no</code> on every citizen; identity model is re-issuable

RISK	LIKELIHOOD	IMPACT	MITIGATION
PDF deed manipulation	Medium	Critical	PAdES signature + SHA-256 stored in DB; verify endpoint compares hash on download
Audit log tampering by privileged user	Low	Critical	INSTEAD OF UPDATE/DELETE triggers at DB level; not a permission, a structural prohibition
Connectivity loss at registration kiosk	High	Low	Wizard state held in browser signals; recovery on reconnect

2.4 Tooling

- **Source control:** Git monorepo ([apps/](#) , [packages/](#) , [infra/](#) , [docs/](#))
- **CI:** GitHub Actions (lint → test → build → docker push)
- **IaC:** Docker Compose for the production stack; nginx for TLS termination
- **Migrations:** Numbered T-SQL files run via [scripts/db/run-migrations.ps1](#) (sqlcmd-based)
- **Diagrams:** Mermaid source rendered to PNG via [mermaid.ink](#)

Requirements Engineering

3.1 Stakeholders

ROLE	PRIMARY CONCERN	SURFACE
Citizen	Register property; carry verifiable ID	Mobile + web
Registry Officer	Review submissions; approve/reject deeds	Officer portal
ID-Issuance Officer	Capture biometrics, encode card	ID-issuer kiosk
Super Admin	User management, system config	Admin console
Public Verifier	Confirm a deed/card is genuine	verify.sarh.ly (no login)
LVCT (operator)	Uptime, security, data sovereignty	Ops dashboards

3.2 Functional Requirements

ID	REQUIREMENT	PRIORITY
FR-1	The system shall authenticate citizens by national ID + password	MUST
FR-2	The system shall allow citizens to register a property by drawing a polygon and uploading documents	MUST
FR-3	The system shall reject overlapping centroids on approved properties (geographic uniqueness)	MUST
FR-4	The system shall warn — but not block — overlapping polygons during officer review	MUST
FR-5	The system shall issue a PAdES-signed PDF deed on approval	MUST
FR-6	The system shall produce a verifiable QR pointing to <code>verify.sarh.ly/{code}</code>	MUST
FR-7	The system shall encode an NTAG 424 DNA card per citizen with rolling-counter SUN URLs	MUST
FR-8	The system shall issue a Verifiable Credential (W3C VC, did:sov) on each approved property	SHOULD

ID	REQUIREMENT	PRIORITY
FR-9	The system shall allow public verification of any deed without login	MUST
FR-10	The system shall maintain a Citizen → Properties timeline visible only to the owning citizen and reviewing officers	MUST
FR-11	The system shall preserve a <code>legacy_national_no</code> reference on every citizen for paper-era continuity	MUST
FR-12	The system shall support Arabic-RTL UI as the default language	MUST
FR-13	The system shall surface a tamper banner if the served deed PDF SHA-256 differs from the stored hash	MUST

3.3 Non-Functional Requirements

ID	CATEGORY	TARGET
NFR-1	Performance — API p95	≤ 250 ms for read endpoints under 100 RPS
NFR-2	Availability	99.5% measured monthly
NFR-3	Security — passwords	bcrypt cost factor 12 minimum
NFR-4	Security — JWT	HS256, 24-hour TTL, no PII in claims
NFR-5	Audit	Every write emits an <code>audit_log</code> row; rows are append-only
NFR-6	Localization	All citizen-facing UI Arabic; error envelope carries <code>message_ar</code> + <code>message_en</code>
NFR-7	Recoverability	RPO ≤ 1 hour, RTO ≤ 4 hours
NFR-8	Data sovereignty	All PII at rest within Libyan-jurisdiction infrastructure
NFR-9	Accessibility	WCAG 2.1 AA for the public verify page
NFR-10	Observability	Structured logs (Serilog) + healthcheck endpoint

3.4 Constraints (CLAUDE.md)

Hard constraints — non-negotiable:

1. Arabic-first RTL for ALL user-facing UI. Latin only for codes, IDs, technical labels.
2. Ownership of digital identity must be re-issuable: when (if) Libya launches a national digital ID, the system must migrate without data loss.
3. Two approved properties cannot share the same centroid; polygon overlap triggers a reviewer warning, not a hard block.
4. Use NTAG 424 DNA only; static UID is never trusted alone — every read must verify a SUN message.
5. Every issued PDF deed must be PAdES-signed and contain a verifiable QR.
6. The audit log is append-only — no UPDATE or DELETE allowed at the database level.
7. No free-text role checking in code — always go through the JSON permission map on `officers.permissions`.

System Architecture

4.1 High-Level Architecture

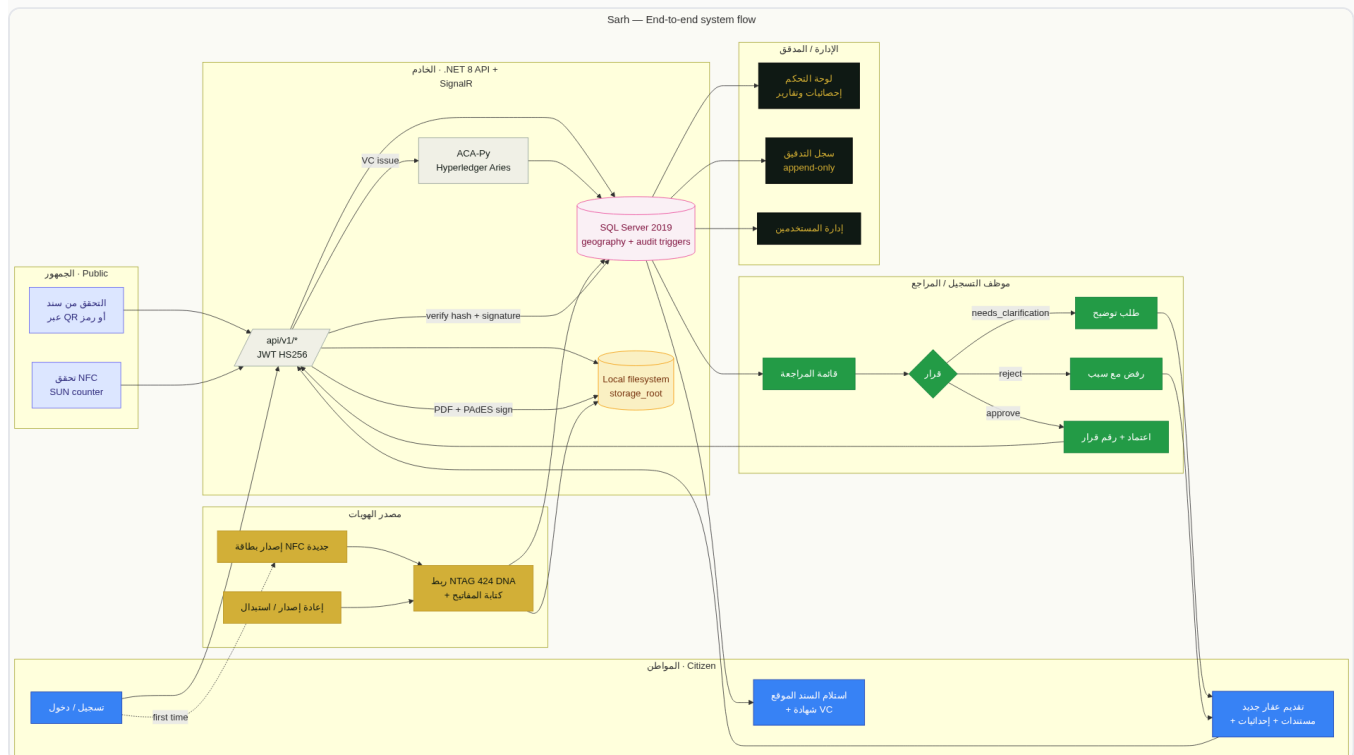


Figure 4.1 — End-to-end system flow across clients, API, datastore, and third-party services.

4.2 Three-Tier Decomposition

PRESENTATION TIER

- Single Angular 21 SPA (`apps/web/`) routed by role to citizen, officer, ID-issuer, admin, and verify shells
- Flutter mobile app (`apps/mobile/`) for citizens — Riverpod, go_router, mapbox
- Standalone components, Angular Signals, `@if / @for` control flow
- Material is *not* used; brand SCSS tokens drive a paper-card aesthetic

APPLICATION TIER

- ASP.NET Core 8 API at `apps/api-dotnet/`
- EF Core 8 for persistence; `Microsoft.Data.SqlClient`
- HS256 JWT signed in `JwtTokenService.cs` ; `[OfficerOnly]` filter for permissioned routes
- Audit interceptor stamps every write

- QuestPDF + QRCode render the deed PDF

DATA TIER

- SQL Server 2019/2022 — local for dev, hosted for prod
- `geography` type for parcel polygons (SRID 4326)
- Numbered T-SQL migrations `000-025` in `infra/mssql/migrations/`
- `INSTEAD OF UPDATE/DELETE` triggers protect `audit_log`
- JSON columns validated by `CHECK ISJSON(col) = 1`

CROSS-CUTTING SERVICES

- Hyperledger ACA-Py (did:sov) for VC issuance
- nginx terminates TLS, enforces CORS, rate-limits `/auth/*`
- ClamAV scans every uploaded document
- Local filesystem at `STORAGE_ROOT` for documents, photos, deeds

4.3 Architecture Decision Records (key choices)

ADR	DECISION	RATIONALE
ADR-001	SQL Server over PostgreSQL/Supabase	Operator preference, Windows-native ops, <code>INSTEAD OF</code> triggers for audit invariants
ADR-002	Single Angular app over four micro-frontends	One deployment, one bundle, role-based shells; the four legacy apps are kept only as a component-migration source
ADR-003	Custom HS256 JWT, not OAuth	No external IdP needed; HS256 + bcrypt is sufficient for the kiosk model, simpler ops
ADR-004	QuestPDF over headless Chromium for deeds	Server-side, deterministic, smaller footprint, PAdES-friendly
ADR-005	NTAG 424 DNA, not generic NFC	SUN message with rolling counter is the only economical anti-clone primitive at card price point
ADR-006	Append-only audit at trigger level, not application level	An attacker with admin rights still cannot rewrite history; structural defense beats permission-based defense

The diagram is a conceptual ERD for a digital ID system. It features the following entities and their attributes:

- REGION**: `code` (string, PK), `name_ar` (string)
- OFFICER**: `role` (string, PK), `region` (string, FK to REGION)
- CITIZEN**: `national_id` (string, PK), `full_name_ar` (string), `region` (string, FK to REGION)
- PROPERTY**: `property_code` (string, PK), `status` (string), `boundary` (polygon)
- DIGITAL_ID_CARD**: `digital_id_number` (string, PK), `status` (string)
- REGISTRATION_REQUEST**: `request_no` (string, PK), `current_status` (string)
- PROPERTY_DOCUMENT**: `doc_type` (string, PK)
- DEED_PDF**: `sha256_hash` (string, PK), `padres_signed` (bool)
- NFC_CARD**: `nfc_uid` (string, PK), `counter` (bigint)
- ID_ISSUANCE_EVENT**: `action` (string, PK)
- NOTIFICATION**: `channel` (string, PK)
- AUDIT_ENTRY**: `id` (bigint, PK), `action` (string)
- SSI_WALLET**: `did` (string, PK)
- VERIFIABLE_CREDENTIAL**: `type` (string, PK)

Key relationships include:

- REGION** to **OFFICER** and **CITIZEN** (scopes, 1:M)
- OFFICER** to **PROPERTY** (owns, 1:M)
- OFFICER** to **DIGITAL_ID_CARD** (issues, 1:M)
- CITIZEN** to **DIGITAL_ID_CARD** (holds, 1:M)
- CITIZEN** to **NOTIFICATION** (receives, 1:M)
- CITIZEN** to **AUDIT_ENTRY** (writes, 1:M)
- CITIZEN** to **SSI_WALLET** (controls, 1:M)
- CITIZEN** to **VERIFIABLE_CREDENTIAL** (stores, 1:M)
- PROPERTY** to **DIGITAL_ID_CARD** (issues, 1:M)
- PROPERTY** to **DEED_PDF** (produces, 1:M)
- DEED_PDF** to **VERIFIABLE_CREDENTIAL** (embeds, 1:M)
- REGISTRATION_REQUEST** to **PROPERTY** (submitted as, 1:M)
- REGISTRATION_REQUEST** to **PROPERTY_DOCUMENT** (reviews, 1:M)
- DIGITAL_ID_CARD** to **NFC_CARD** (encoded as, 1:M)
- DIGITAL_ID_CARD** to **ID_ISSUANCE_EVENT** (tracked by, 1:M)

5.2 Logical ERD



5.4 Conventions

- Snake_case table and column names
- Primary keys: `UNIQUEIDENTIFIER DEFAULT NEWID()` — except `audit_log.id` which is `BIGINT IDENTITY` for monotonic ordering
- Time columns: `DATETIMEOFFSET(3) DEFAULT SYSDATETIMEOFFSET()`
- Auto-stamped `updated_at` via `AFTER UPDATE` trigger on every entity
- Soft delete: `is_active = 0` — never physical delete
- Geometry: SQL Server `geography` (SRID 4326 implicit). Spatial methods used: `STIntersects`, `STDistance`, `EnvelopeCenter()`, `STArea()`
- ENUMs: SQL Server has none — modeled as `NVARCHAR(N) CHECK (col IN (...))`
- JSONB equivalent: `NVARCHAR(MAX)` with `CHECK ISJSON(col) = 1`

5.5 Append-Only Audit (the structural defense)

```
CREATE TRIGGER trg_audit_log_no_update ON audit_log
INSTEAD OF UPDATE AS
BEGIN
    RAISERROR ('audit_log is append-only', 16, 1);
END;

CREATE TRIGGER trg_audit_log_no_delete ON audit_log
INSTEAD OF DELETE AS
BEGIN
    RAISERROR ('audit_log is append-only', 16, 1);
END;
```

The trigger is the contract: even a privileged DBA cannot rewrite history through application code paths without first dropping the trigger — an action which is itself logged at the SQL Server engine level.

Class Diagram

The class diagram captures the principal domain types in the .NET API layer and the relationships between them. Each class corresponds either to a database entity or to a service that orchestrates writes.

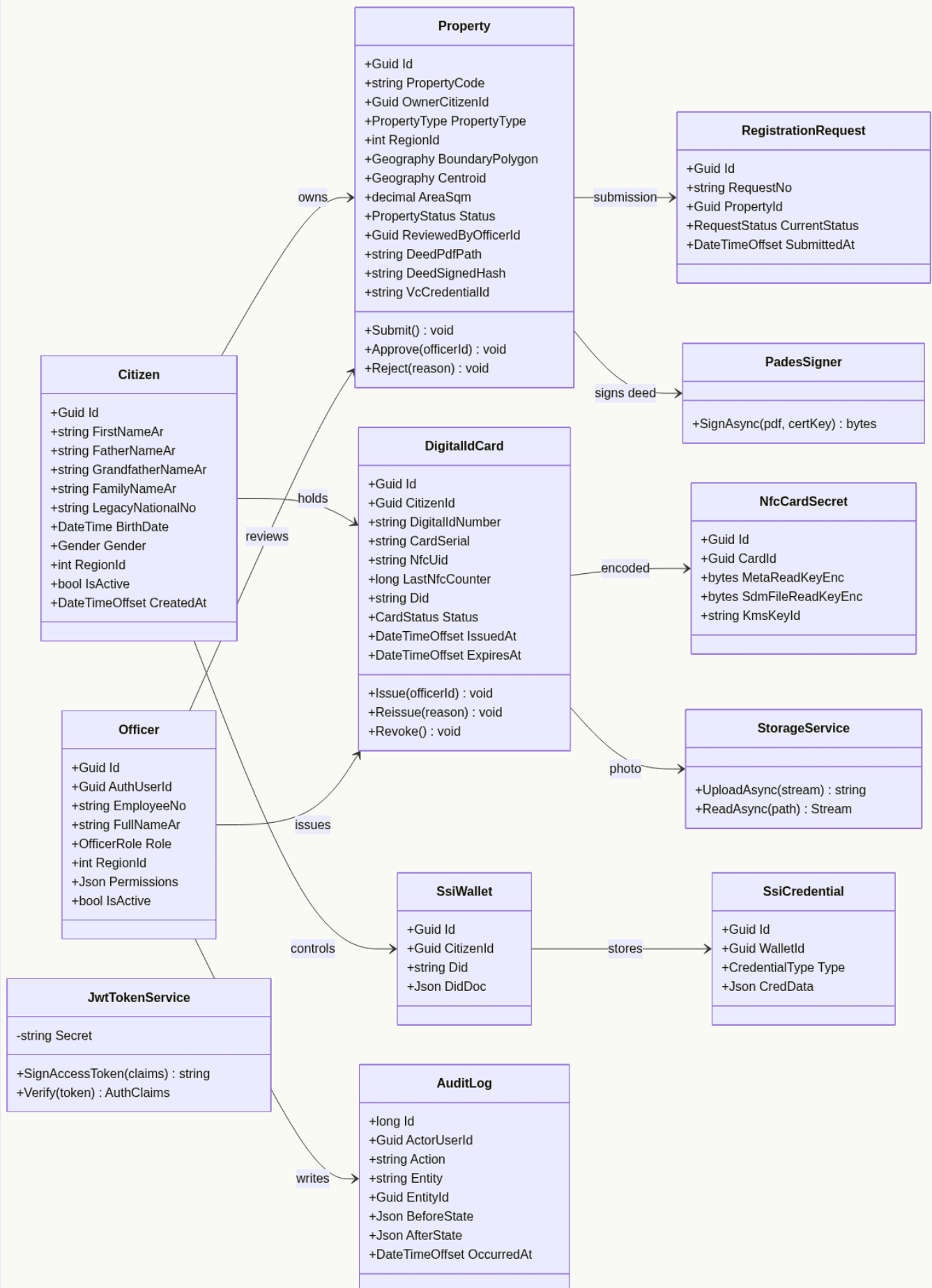


Figure 6.1 — Principal domain classes and their relationships.

6.1 Service Layer Highlights

- `JwtTokenService` — signs and validates HS256 tokens from `SARH_JWT_SECRET`.
- `StorageService` — abstracts the local filesystem under `STORAGE_ROOT`; all upload paths go through this single seam.
- `ReviewService` — orchestrates officer approval: persists state, invokes `DeedPdfBuilder`, computes SHA-256, writes to `audit_log`.
- `DeedPdfBuilder` — QuestPDF + QRCoder; renders A4 Arabic-RTL deed with three card sections, parcel map, and verify QR.
- `VerifyService` — public, no-auth resolver from deed code → public deed view (incl. SHA-256 for client integrity check).

Sequence Diagrams

7.1 Sign-In

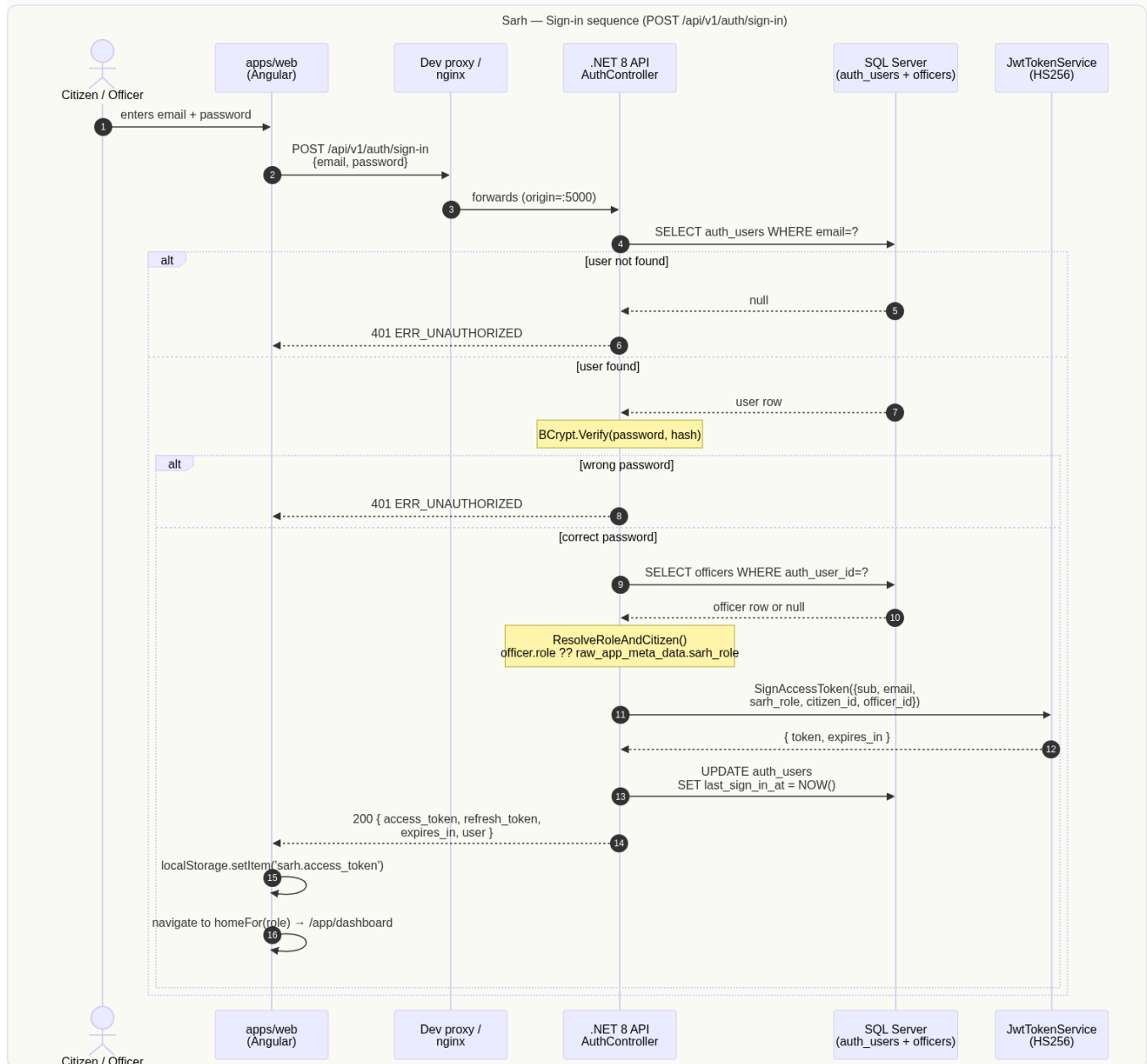


Figure 7.1 — Citizen sign-in: bcrypt verify + JWT issuance.

7.2 ID Issuance (5-step kiosk wizard)

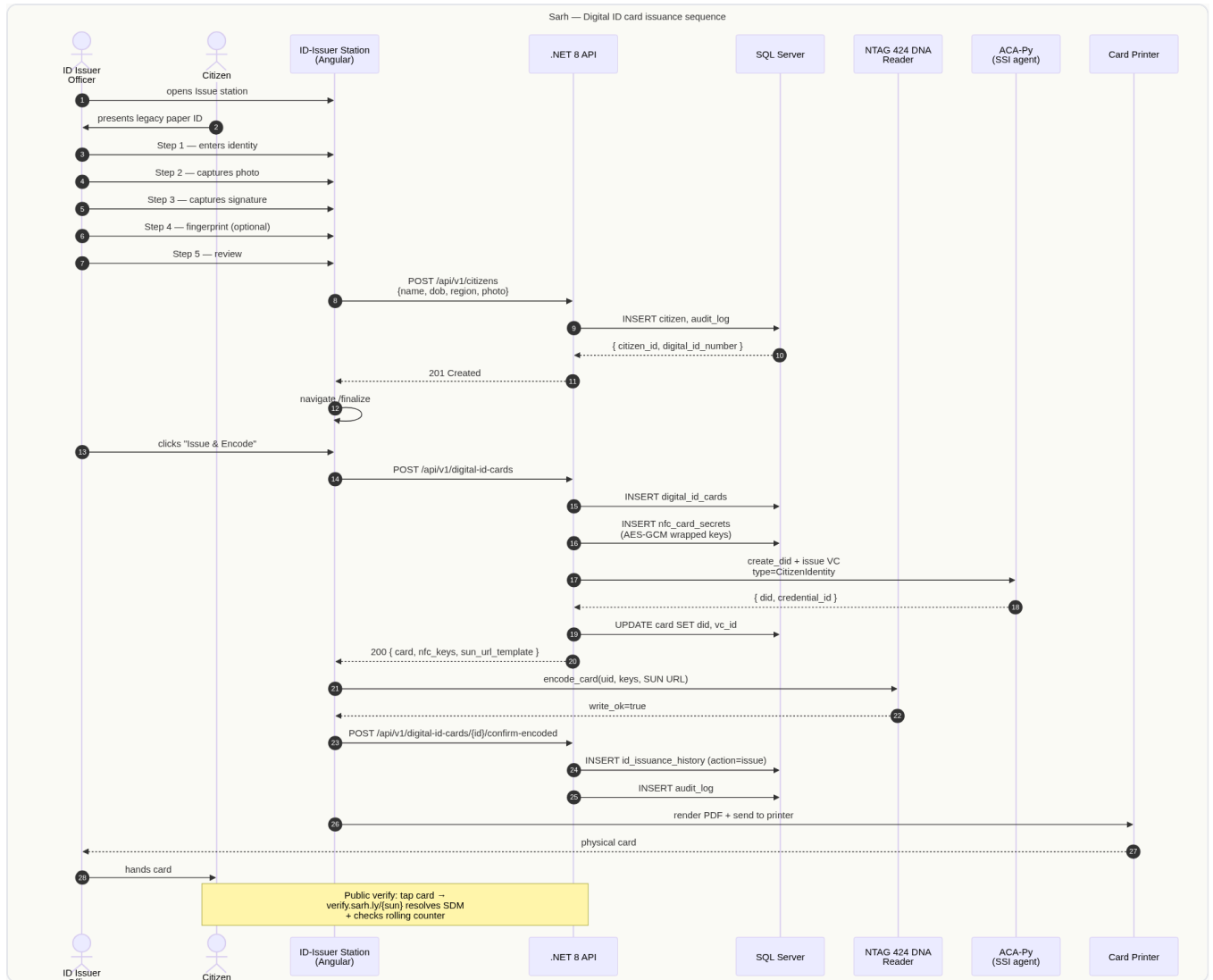


Figure 7.2 — End-to-end ID issuance: identity capture, photo, signature, fingerprint, NFC encoding.

7.3 Property Approval

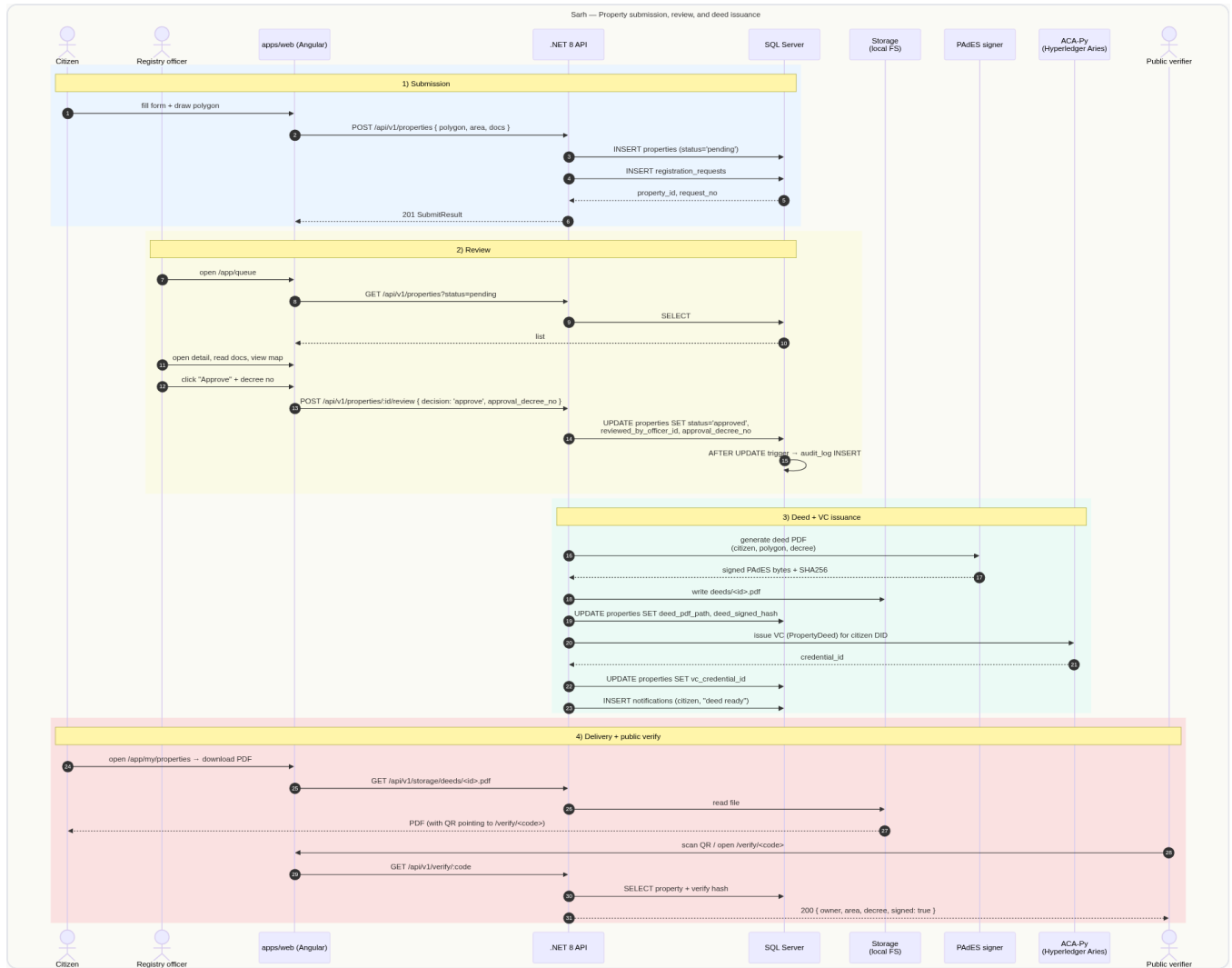
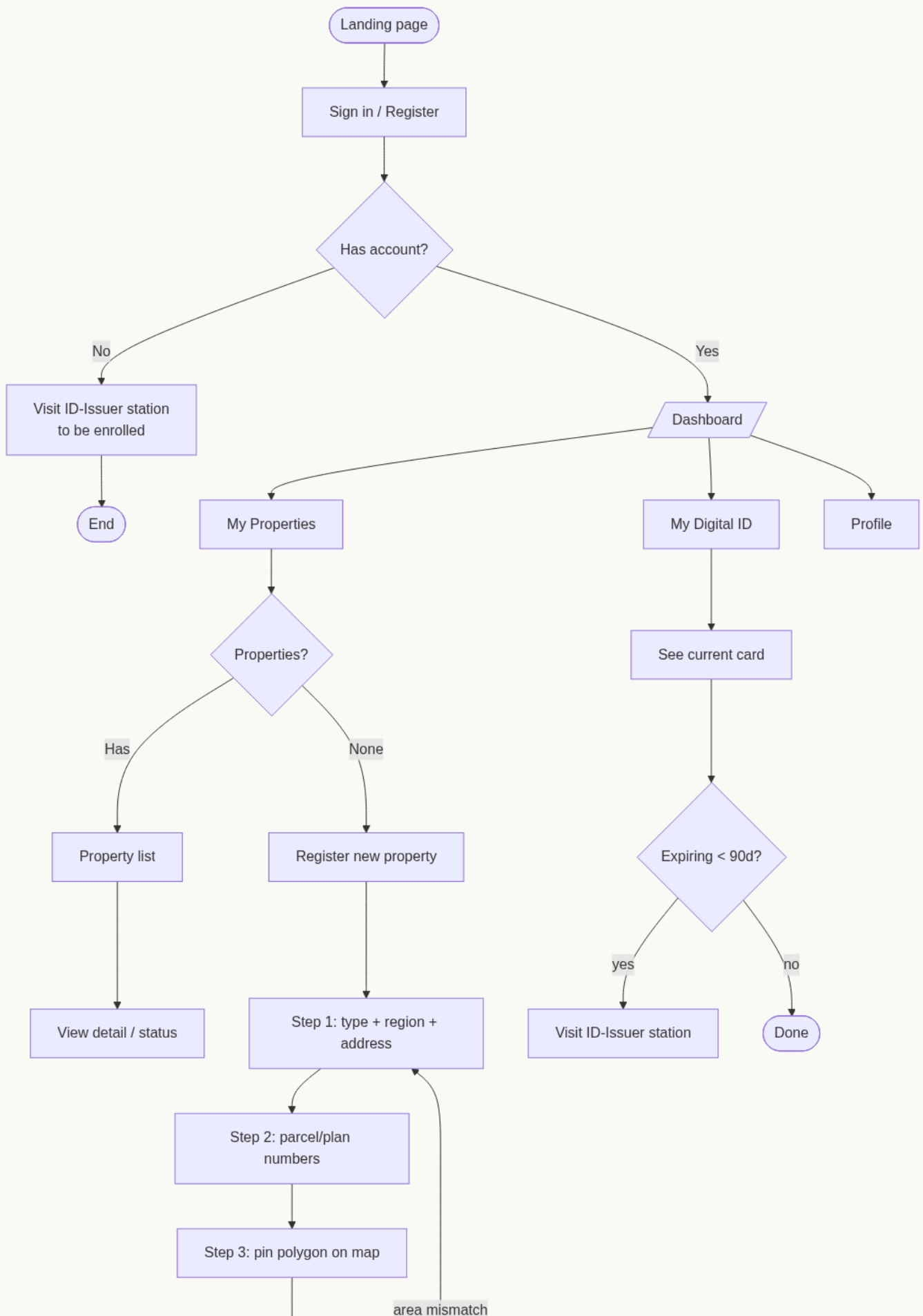


Figure 7.3 — Officer review path: validation, polygon overlap warning, deed PDF generation, audit log write.

User Flows

8.1 Citizen Flow

Sarh — Citizen user flow



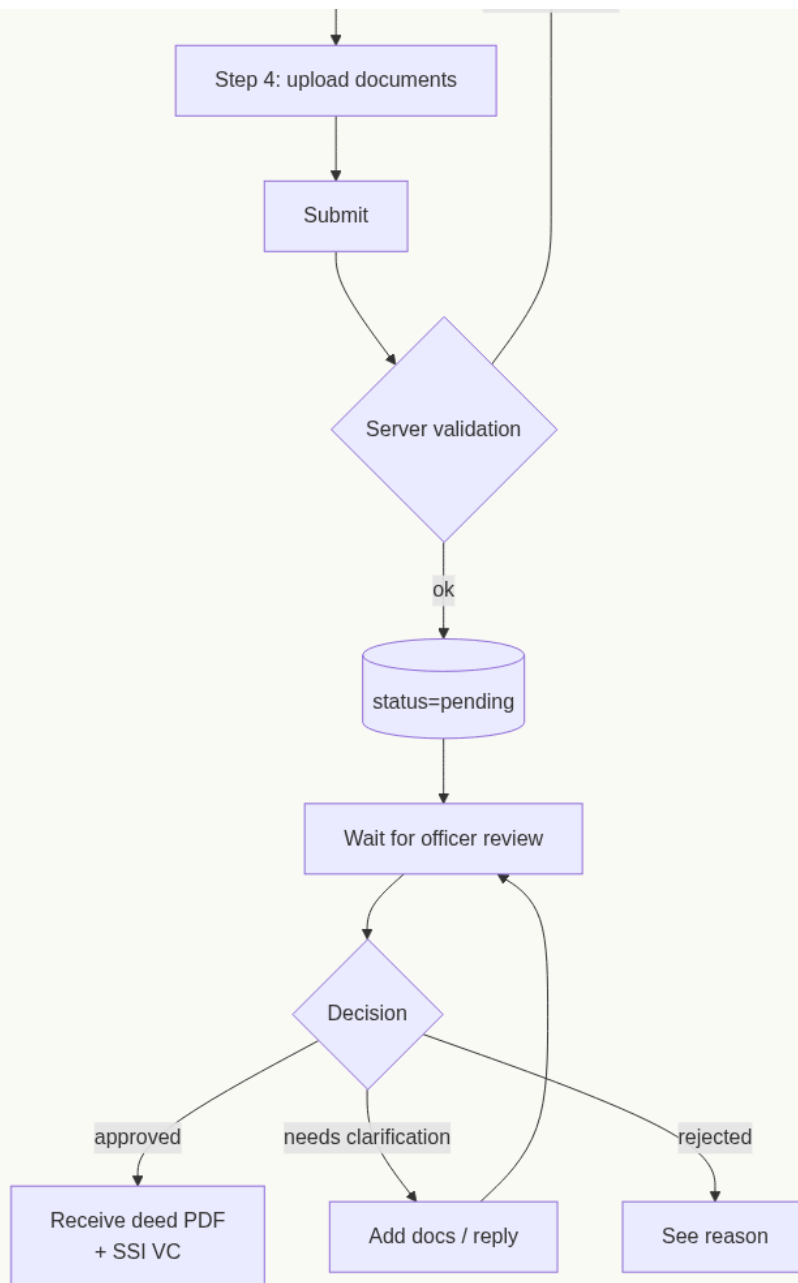


Figure 8.1 — Citizen journey from sign-in through property submission and deed download.

8.2 Officer Flow

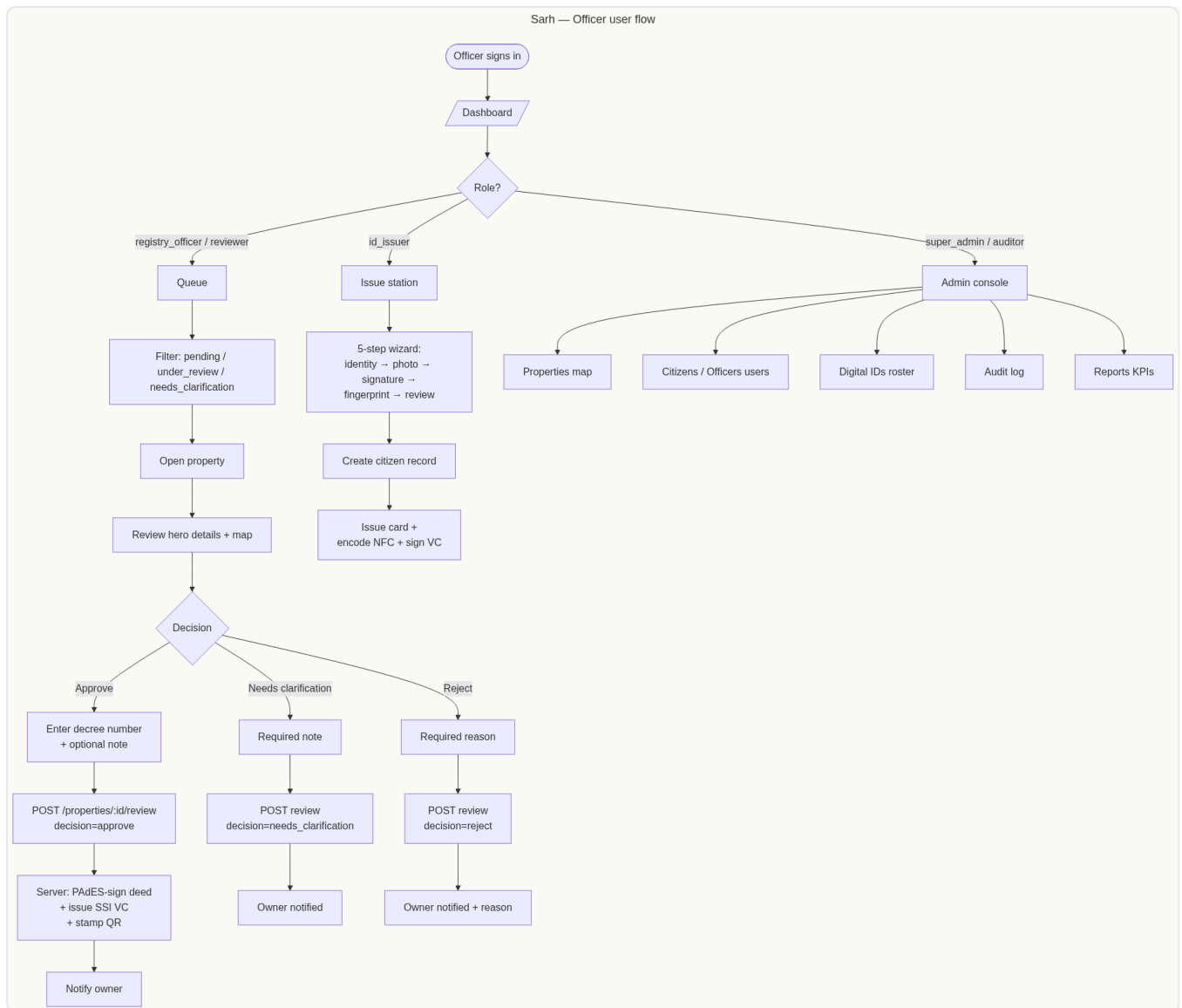


Figure 8.2 — Officer journey through queue → review → approve / reject decisions.

UI/UX Design

9.1 Design Philosophy

The visual system is anchored in a "PropTech paper-card" aesthetic: warm paper background, slate ink, deliberate orange accents for action, cyan for confirmation, red for caution. We deliberately avoid green to keep the language of "approved" anchored in a neutral cool tone (cyan), which reads as *verified by the system* rather than *positive emotion*.

9.2 Color Tokens

Primary — Slate
#0F172A
Headings, ink, sidebar

Accent — Orange
#F97316
Primary action, focus

Good — Cyan
#0891B2
Approved, ✓ state

Warn — Red
#DC2626
Errors, rejections

Paper
#FAFAF9
Background canvas

Muted
#64748B
Secondary text

9.3 Typography

SURFACE	FAMILY	NOTES
Arabic UI	IBM Plex Arabic, Cairo, Tajawal	RTL, font-feature-settings opt for ligatures
Latin UI	Inter, Segoe UI, system-ui	Tabular numerals on for IDs and counts
Code / IDs	JetBrains Mono	For citizen national_no, deed codes, hashes

9.4 Layout & Spacing

- Sidebar 240 px collapsed shell, fixed at top breakpoint
- Card radius 14 px, border 1 px `--rule`, padding 22 px
- Form rows 14 px gap; section dividers 1 px `--rule`
- Buttons 9 px × 18 px, font-weight 700, transition 120 ms

9.5 Internationalization

Default direction is RTL. Logical CSS properties (`margin-inline-end` , `padding-inline-start`) are used throughout so the same stylesheet flips correctly when an LTR shell (e.g. the verify page) is rendered. Validation messages, placeholders, button labels, and toast copy are all Arabic by default; the verify page additionally exposes English to support foreign banks and courts.

Wireframes

Five low-fidelity wireframes establish the page-level grammar of the system. They are intentionally sparse: skeleton structure, no microcopy, no real iconography. Their purpose is to fix layout and information hierarchy before higher-fidelity work begins.

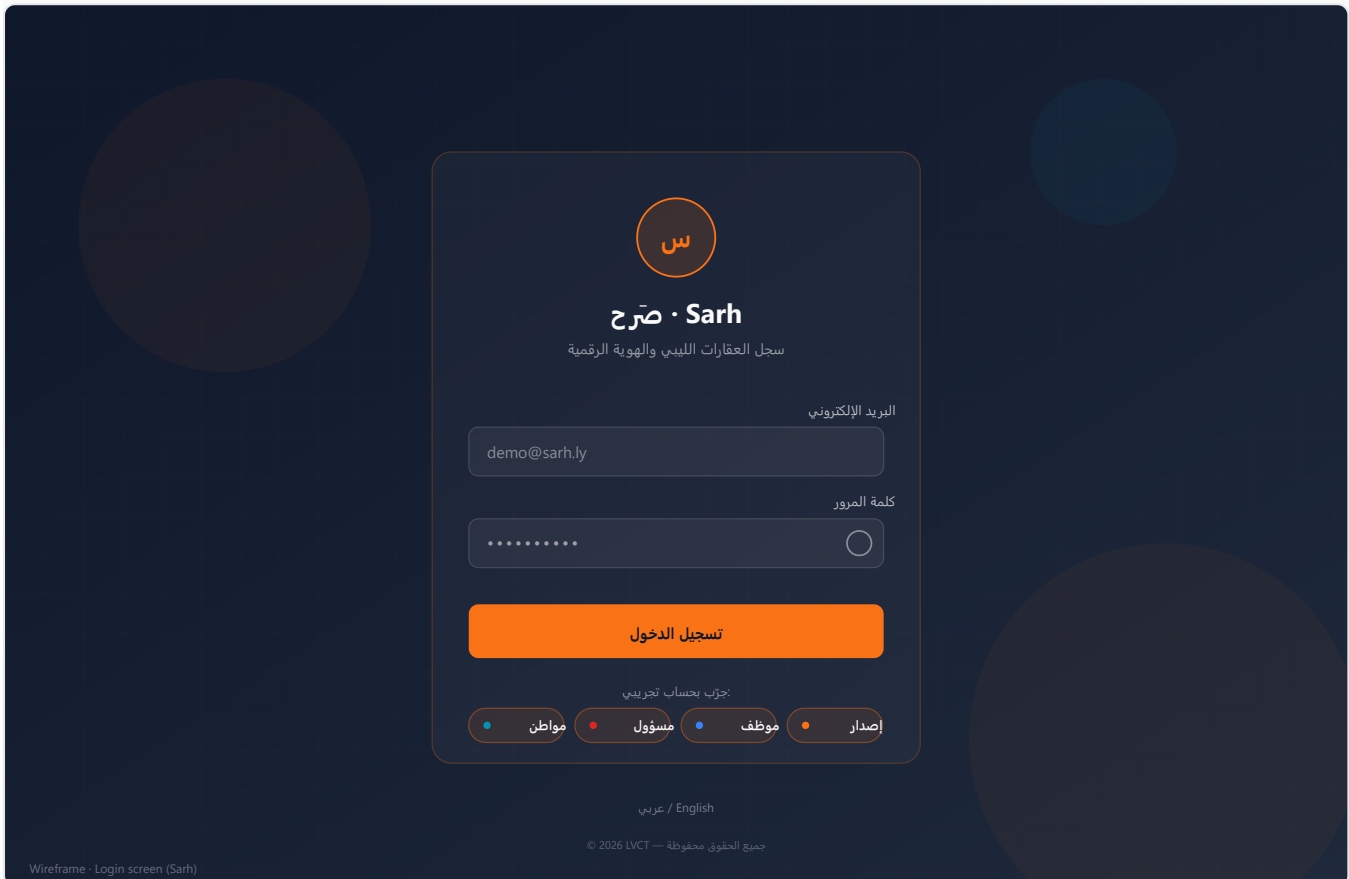


Figure 10.1 — Login screen: split-pane with brand panel and form.

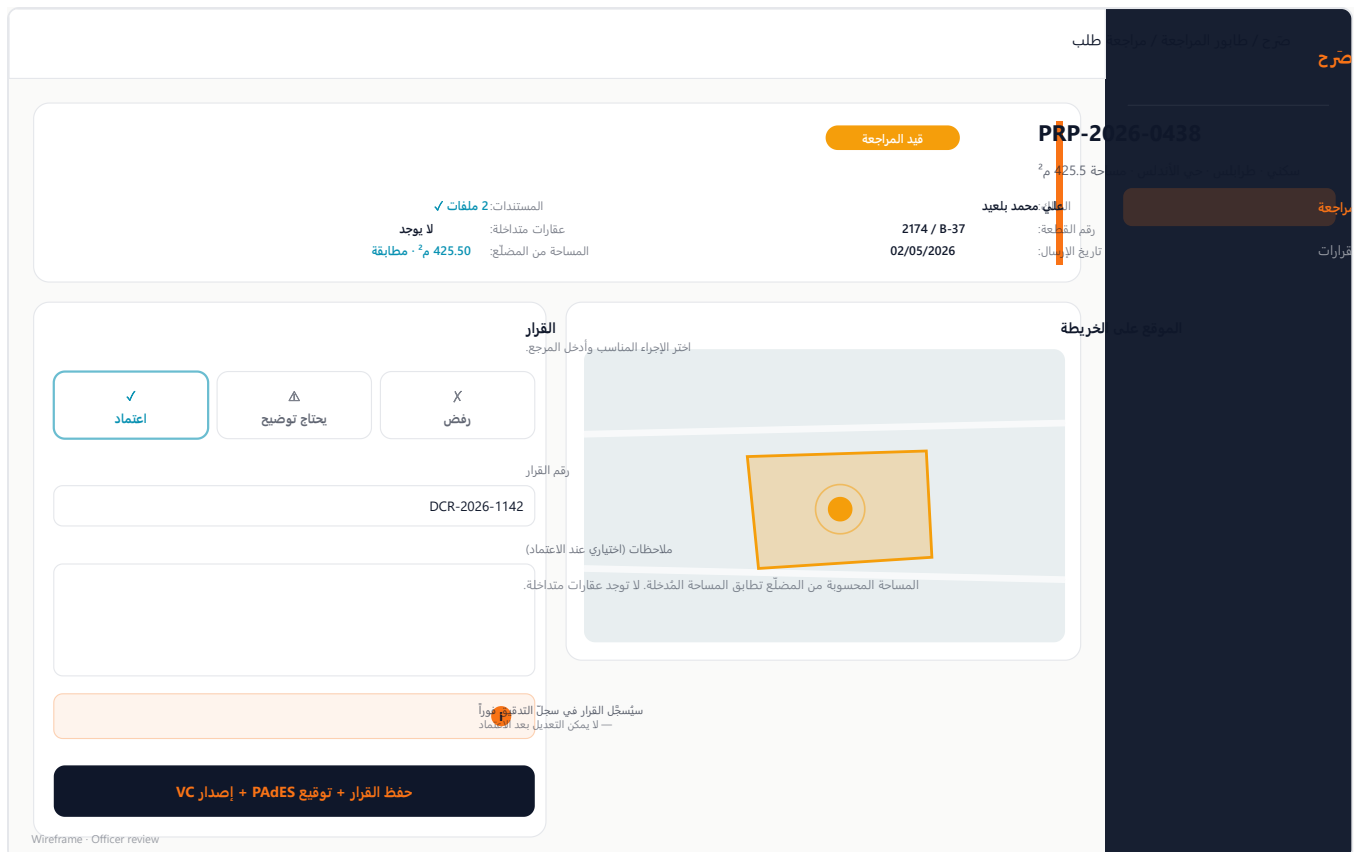


Figure 10.4 — Officer review: queue + map preview + decision panel.

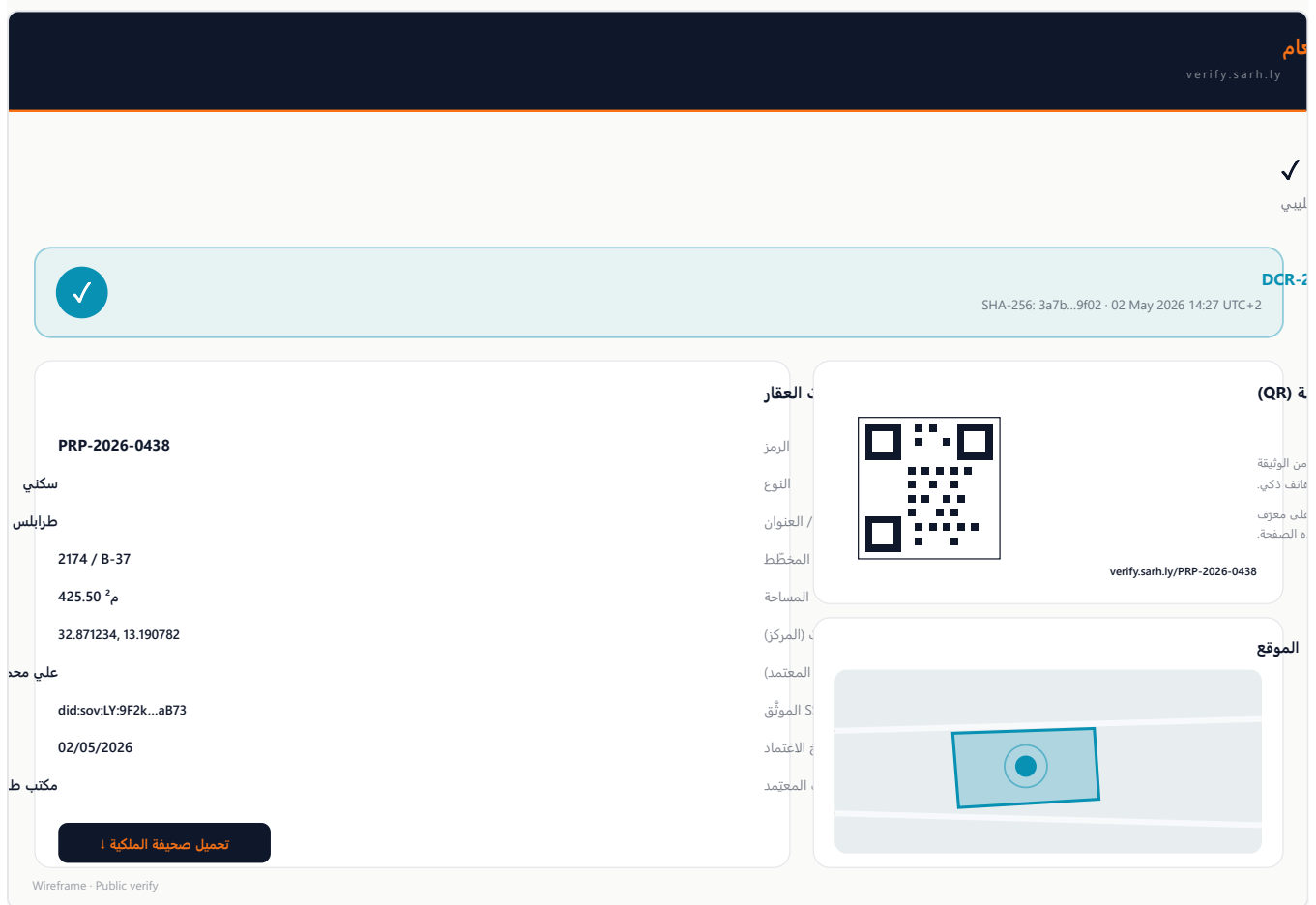


Figure 10.5 — Public verify: tamper banner + deed metadata + QR + downloadable PDF.

API Contracts

11.1 Conventions

- Base path: `/api/v1/`
- Resource names: English plural — `/citizens` , `/properties` , `/digital-id-cards`
- Pagination: cursor-based — `?cursor=&limit=20` — never offset on large tables
- Error envelope: `{ "error": { "code": "...", "message_ar": "...", "message_en": "..." } }`
- Every write endpoint emits an `audit_log` row through a global filter

11.2 Selected Endpoints

METHOD	PATH	AUTH	DESCRIPTION
POST	<code>/auth/login</code>	Public	Citizen or officer sign-in; returns JWT
GET	<code>/citizens/me</code>	Citizen	Self-profile + linked properties summary
POST	<code>/properties</code>	Citizen	Submit a property registration with polygon + documents
GET	<code>/properties/queue</code>	Officer	Pending submissions in officer's region
POST	<code>/properties/{id}/approve</code>	Officer	Approve, render PDF deed, store hash
POST	<code>/properties/{id}/reject</code>	Officer	Reject with Arabic reason
POST	<code>/digital-id-cards</code>	ID-Issuer	Capture biometrics, encode NTAG 424 DNA card
GET	<code>/verify/{code}</code>	Public	Public deed view with hash for tamper check
GET	<code>/verify/{code}/deed.pdf</code>	Public	Streams PDF; sets <code>X-Deed-SHA256</code> ; returns 409 on mismatch

11.3 Error Codes (selected)

CODE	HTTP	MEANING
<code>ERR_AUTH_INVALID</code>	401	Bad credentials

CODE	HTTP	MEANING
ERR_PERMISSION	403	Permission map denies the action
ERR_GEO_CENTROID_DUP	409	Approved property already exists at the same centroid
ERR_GEO_OVERLAP_WARN	200 (warning)	Polygon overlaps an existing one — review required, not blocked
ERR_DEED_HASH_MISMATCH	409	Stored SHA-256 does not match served PDF — tamper indicator
ERR_NFC_SUN_INVALID	400	SUN counter outside accepted window

Security & Compliance

12.1 Trust Boundaries

1. **Public** ↔ **API**: TLS 1.2+ at nginx; rate-limited `/auth/*` and `/properties`
2. **API** ↔ **DB**: Mutual auth via `sarh_app` SQL login with least-privilege grants; no `db_owner`
3. **API** ↔ **ACA-Py**: Internal Docker network only; admin port not exposed
4. **API** ↔ **Storage**: Single seam via `StorageService`; path traversal blocked at boundary

12.2 Authentication & Authorization

- JWT HS256, signing key = `SARH_JWT_SECRET`
- Claims: `sub`, `citizen_id` or `officer_id`, `role`, `permissions` (JSON), `exp`
- No PII in claims — never email, name, or national number
- Officers carry a JSON permission map; `[OfficerOnly("permission.key")]` filter checks the map without DB roundtrip
- Passwords: bcrypt cost ≥ 12 (BCrypt.Net-Next)

12.3 Tamper Evidence

Every approved deed PDF is rendered by `DeedPdfBuilder` (QuestPDF + QRCoder), then SHA-256-hashed by `ReviewService`; the hash and the on-disk path are stored on `properties.deed_signed_hash` and `properties.deed_pdf_path`. On every public download via `GET /verify/{code}/deed.pdf`, the server reads the file, recomputes SHA-256, compares against the stored hash, and returns HTTP 409 if they diverge — this is the system's primary tamper indicator. The hash is also surfaced on the public verify page so any third party can re-verify it against their own download.

12.4 NFC Anti-Clone

NTAG 424 DNA chips emit a Secure Unique Number (SUN) message at every read — a URL with a rolling counter and a per-card MAC. The verification server rejects a SUN whose counter is below the last-seen value (replay) or whose MAC does not validate against the per-card key. A static UID alone is never treated as proof of authenticity.

12.5 Audit Log

Every write endpoint emits a row into `audit_log`. Rows carry actor, action, target entity, before-state, after-state, IP address, and timestamp. The table has `INSTEAD OF UPDATE` and `INSTEAD OF DELETE` triggers that raise an error — no application code path can rewrite history. Trigger removal would be visible in SQL Server's own DDL audit.

12.6 Compliance Checklist

- All citizen-facing surfaces serve Arabic by default
- Storage buckets / directories private by default; ACL on every read
- JWT contains no PII
- No secrets in repository — `.env.local` is gitignored
- CORS restricted to known frontend origins
- Rate limit on `/auth/*` and `/properties/submit`
- ClamAV scan on every uploaded document

Roadmap & Future Work

13.1 Delivery Phases

PHASE	TITLE	STATUS
0	Repo scaffold + infra	Complete
1	Schema + RLS-equivalent + seed	Complete
2	Backend auth + citizens module	Complete
3	Digital ID issuance + NFC encoding	Complete
4	Properties module + spatial queries	Complete
5	Workflow + officer review + notifications	Complete
6	SSI wallet + VC issuance	In progress
7	Citizen mobile app (Flutter)	In progress
8	Officer web portal (Angular)	Complete
9	ID-issuer station (Angular + NFC + camera)	Complete
10	Admin console + reports	Complete
11	Public verification site	Complete
12	Hardening, audit, deployment	Pending

13.2 Future Work

- **Federation with banks:** presentation requests for property VCs at loan origination
- **Court e-filing:** dispute resolution that consumes audit log evidence
- **National eID launch path:** migrate from `legacy_national_no` to the new state identifier without data loss
- **Cadastral survey ingest:** bulk import of officially surveyed parcels
- **Selective disclosure:** ZK-proofs over property attributes (age > threshold without revealing area)

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